Estimating Regression Coefficients
(the slope and intercept of regression line)
using Gradient Descent Algorithm

What is Gradient Descent?
Gradient descent is an optimization algorithm used in machine learning to minimize the cost function by iteratively adjusting parameters in the direction of the negative gradient, aiming to find the optimal set of parameters.

Cost function(Loss function): The cost function represents the discrepancy(error) between the predicted output of the model and the actual output. The goal of gradient descent is to find the set of parameters that minimizes this discrepancy and improves the model's performance.

1. The regression model

For simple linear regression, the line is: $\hat{y} = b_0 + b_1 x$

Where b_0 : y-intercept, b_1 : slope

2. Cost function (loss function) (what we minimize)

We measure how wrong our line is using **Mean Squared Error**:

L = MSE = $\sum_{i=1}^{n}(\widehat{y_i} - y_i)^2 = \frac{1}{2n}\sum_{i=1}^{n}(b_0 + b_1 x_i - y_i)^2$

n**:** Total number of data points.

y_i**:** The actual value.

$\widehat{y_i}$**:** The predicted value from the line.

3. Simple algorithm

a. Initialize b_0 , b_1 (e.g., (0, 0))

b. Repeat:

Compute predictions

Compute gradients

Update parameters

c. Stop when changes are very small.

4.

- α: learning rate (step size)
- Small α → slow but stable
- Large α → fast but may overshoot

Question:

Consider the following sample dataset

X	Y
2	3
3	6
4	7

Estimate the regression coefficients b_0 and b_1 by using Gradient Descent Algorithm where learning rate $\alpha = 0.05$. Perform three iteration and compute Loss function (Mean Square Error (MSE)) at each iteration as well. Initialize regression coefficients with (0,0)

Solution:

$$Estimated\ Regression\ Line\ Equation:\ \ \hat{y} = b_0 + b_1 x$$

$$\text{L = MSE = } \sum_{i=1}^{n}(\widehat{y}_i - y_i)^2 = \frac{1}{2n}\sum_{i=1}^{n}(b_0 + b_1 x_i - y_i)^2$$

$$b_0^{new} = b_0^{old} - \alpha\frac{\partial L}{\partial b_0} \qquad ------equation1$$

$$b_1^{new} = b_1^{old} - \alpha\frac{\partial L}{\partial b_1} \qquad ------equation2$$

$$\frac{\partial L}{\partial b_0} = \frac{1}{n}\sum_{1}^{n}(b_0 + b_1 x - y) = \frac{1}{3}(b_0 + 2b_1 - 3 + b_0 + 3b_1 - 6 + b_0 + 4b_1 - 7)$$

$$\frac{\partial L}{\partial b_0} = \frac{1}{3}(3b_0 + 9b_1 - 16) \qquad ------equation3$$

$$\frac{\partial L}{\partial b_1} = \frac{1}{n}\sum_{1}^{n}(b_0 + b_1 x - y)x = \frac{1}{3}(2(b_0 + 2b_1 - 3) + 3(b_0 + 3b_1 - 6) + 4(b_0 + 4b_1 - 7))$$

$$\frac{\partial L}{\partial b_1} = \frac{1}{3}(2(b_0 + 2b_1 - 3) + 3(b_0 + 3b_1 - 6) + 4(b_0 + 4b_1 - 7))$$

$$\frac{\partial L}{\partial b_1} = \frac{1}{3}(9b_0 + 29b_1 - 52) \quad ------equation4$$

Put values of equation 3 and 4 in equation 1 and 2,

$$b_0^{new} = b_0^{old} - \ 0.05\frac{1}{3}\left(3b_0^{old} + 9b_1^{old} - 16\right)$$

$$b_1^{new} = b_1^{old} - \ 0.05\ \tfrac{1}{3}\left(9b_0^{old} + 29b_1^{old} - 52\right)$$

Iteration 1:

$$b_0^{(1)} = b_0^{(0)} - \ 0.05\frac{1}{3}\left(3b_0^{(0)} + 9b_1^{(0)} - 16\right) = 0 - \ 0.05\frac{1}{3}(3(0) + 9(0) - 16) = 0.26667$$

$$b_1^{(1)} = b_1^{(0)} - \ 0.05\ \tfrac{1}{3}\left(9b_0^{(0)} + 29b_1^{(0)} - 52\right) = 0 - \ 0.05\ \tfrac{1}{3}(9(0) + 29(0) - 52) = 0.86667$$

$$L = \frac{1}{6}((0.26667 + 2 * 0.86667 - 3)^2 + (0.26667 + 3 * 0.86667 - 6)^2 + (0.26667 + 4 * 0.86667 - 7)^2)$$

$$L = 3.58145$$

Iteration 2:

$$b_0^{(2)} = b_0^{(1)} - \ 0.05\frac{1}{3}\left(3b_0^{(1)} + 9b_1^{(1)} - 16\right) = 0.26667 - \ 0.05\frac{1}{3}(3(0.26667) + 9(0.86667) - 16)$$
$$= 0.39$$

$$b_1^{(2)} = b_1^{(1)} - \ 0.05\ \tfrac{1}{3}\left(9b_0^{(1)} + 29b_1^{(1)} - 52\right)$$

$$= 0.86667 - \ 0.05\ \frac{1}{3}(9(0.26667) + 29(0.86667) - 52) = 1.27445$$

$$L = \frac{1}{6}((0.39 + 2 * 1.27445 - 3)^2 + (0.39 + 3 * 1.27445 - 6)^2 + (0.39 + 4 * 1.27445 - 7)^2)$$

$$L = 0.91377$$

Iteration 3:

$$b_0^{(3)} = b_0^{(2)} - \ 0.05\frac{1}{3}\left(3b_0^{(2)} + 9b_1^{(2)} - 16\right) = 0.39 - \ 0.05\frac{1}{3}(3(0.39) + 9(1.27445) - 16) = 0.446$$

$$b_1^{(3)} = b_1^{(2)} - \ 0.05\ \tfrac{1}{3}\left(9b_0^{(2)} + 29b_1^{(2)} - 52\right)$$

$$= 1.27445 - \ 0.05\ \frac{1}{3}(9(0.39) + 29(1.27445) - 52) = 1.46663$$

$$L = \frac{1}{6}((0.446 + 2 * 1.46663 - 3)^2 + (0.446 + 3 * 1.46663 - 6)^2 + (0.446 + 4 * 1.46663 - 7)^2)$$

$$L = 0.32474$$